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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 02/05/2015 |
| **PI Name:** | Yu, Nan Ph.D. | | |
| **Project Title:** | JPL Participation in ESA ACES (Atomic Clock Ensemble in Space) Worldwide Clock Comparison Campaign | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Fundamental physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | NASA CENTER | **Phone:** | |
| **Organization Name:** | NASA Jet Propulsion Laboratory | | |
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| **City:** | Pasadena | **State:** | CA |
| **Zip Code:** | 91109-8001 | **Congressional District:** | 27 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2010 Fundamental Physics Rapid Turn Around Flight Opportunity |
| **Start Date:** | 10/01/2010 | **End Date:** | 03/31/2018 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | Callas, John | **Contact Phone:** | |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 3/31/2018 per PI (Ed., 2/19/15) | | |
| **Key Personnel Changes/Previous PI:** | In addition to PI/Co-I, the following people have made contributions to the task in FY14: William Diener, Fang Zhong, Wade Rellergert, and Slava Turyshev. | | |
| **COI Name (Institution):** | Burt, Eric ( Jet Propulsion Laboratory )<br>Prestage, John ( Jet Propulsion Laboratory )<br>Tjoelker, Robert ( Jet Propulsion Laboratory ) | | |
| **Grant/Contract No.:** | JPL Task Plan number 71-15743 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Tremendous progress has been made recently in the U.S. and other countries in advancing atomic clock technology and its applications for the benefit of science, national interests, and society in general. Space provides a unique environment in which many benefits can be realized. The European Space Agency (ESA) is readying the ACES (Atomic Clock Ensemble in Space) payload for the International Space Station (ISS) for an anticipated 2013 launch. Some of the key objectives of ACES will be achieved by global time transfer and clock comparisons with the most advanced clocks at multiple worldwide sites. There is an opportunity for investigators from the leading U.S. clock and metrology labs to participate in the clock comparison campaign, enhancing the overall science objective of ACES and at the same time demonstrating and providing science benefit from advanced U.S. clock technology.<br>The Jet Propulsion Laboratory (JPL) is one of the major U.S. time and frequency metrology institutions serving NASA missions and Deep Space Network needs. It was the NASA center for the microgravity LCAP program (Laser Cooling and Atomic Physics) in charge of developing the cold atom clock payload for PARCS (Primary Atomic Reference Clock in Space). JPL has also developed a unique atomic frequency standard that has demonstrated an exceptional long-term stability. We propose to collaborate with ESA on the ACES mission by participating in the ACES global clock campaign with JPL's ultra-stable mercury ion clock. During a proposed six-year performance period, we will: 1) contribute significantly to the ACES mission science objectives by providing unique clock references from a Western U.S. site; 2) Provide and demonstrate a high-impact application for JPL's high performance time keeping clock technology; and 3) re-establish NASA's involvement in ISS Clock Science and initiate a healthy collaboration between NASA and ESA in the area of clock and quantum sensors.<br><br>The task is to collaborate with ESA on the ACES mission by participating in the ACES global clock comparison campaign and provide frequency and time references derived from JPL's ultra-stable mercury ion clock. These references will be connected to a microwave link ground terminal provided by ACES and installed at JPL. Clock comparisons will be carried out over the microwave link with ESA clocks on the ISS. As JPL is the only ground comparison site located in the western USA and the mercury ion clock is the only one of its kind in the global clock comparison campaign, JPL's participation will contribute significantly to the overall ACES mission science objectives, while simultaneously providing a high-impact application for JPL's high performance time keeping clock technology. The task will also help re-establish NASA's involvement in ISS clock science and enable future collaboration between NASA and ESA in the area of clocks and quantum sensors. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Advanced atomic clocks are critical to the interests of NASA and the nation. Telecommunication and navigation (e.g., GPS) are well-known applications of space clocks. Within NASA, high performance atomic clocks and frequency and timing capabilities enable operation of the NASA Deep Space Network and provide fundamental tools for precise measurements exploring the laws of Nature. Progress in advanced clock technology development, both on ground and for space, is rapidly being made in many countries. Participation by the U.S. in the ACES mission contributes to the science objectives of the ISS consistent with the 2011 National Research Council (NRC) report recommendations in Fundamental physics and demonstrates the capability and potential of NASA mercury ion clock technology. |
| **Task Progress:** | This is the 4th year of the multi-year task. The first year involved formally establishing the collaboration, beginning needed updates of a JPL ground ion clock, and preparation for the installation of the ESA provided microwave link ground terminal at a JPL site. The second year focused on LITS-9 upgrade, its diagnosis, and the design of the MWL ground terminal site. In the 3rd year, the ground clock development focus has been shifted to leveraging the new microwave ground clock development efforts. The construction of the ground terminal site has mostly been completed this fiscal year. This year, much progress has been made in advancing the completion of the next generation Hg+ ion ground clock with the first version completed and delivered to the sponsor. The ground terminal site construction has been completed and link stability test has been made. In addition, efforts on the science study of the clock link requirements and analysis continued. Our initial finding suggests the overall consistency of the link and orbital determination requirements with minor variations to be further investigated. |
| **Bibliography Type:** | Description: (Last Updated: 10/28/2020) |
| **Abstracts for Journals and Proceedings** | Yu N. "Miniature Optical Clock of Single Trapped Yb+ Ions." Presented at the ESA Workshop on Frequency Standards with Trapped Ions, Neuchatel, Switzerland, June, 2014.<br>Workshop on Frequency Standards with Trapped Ions, Neuchatel, Switzerland, June, 2014. Information at: http://www.ptb.de/emrp/ion-clocks-home.html ; accessed 2/10/15. , Jun-2014 |
| **Abstracts for Journals and Proceedings** | Sullivan S, Rellergert W, Grudinin I, Baumgartel L, Yu N. "Progress Towards a Compact Optical Clock at JPL." 45th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics, Madison, WI, June 2-6, 2014.<br>Bulletin of the American Physical Society. 2014;59(8):BAPS.2014.DAMOP.Q1.67.<br>http://meetings.aps.org/link/BAPS.2014.DAMOP.Q1.67 ; accessed 2/10/15. , Jun-2014 |
| **Papers from Meeting Proceedings** | Burt E, Tucker B, Larsen K, Hamell R, Tjoelker R. "Next Generation JPL Ultra-Stable Trapped Ion Atomic Clocks." Precise Time and Time Interval (PTTI) Meeting, Bellevue, WA, December 2-5, 2013.<br>Proceedings of the 45th Annual Precise Time and Time Interval Systems and Applications Meeting. Bellevue, Washington, December 2013, p. 55-61. http://www.ion.org/publications/abstract.cfm?jp=p&articleID=11572 ; accessed 2/10/15. , Dec-2013 |